SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Brungi Shiva Ganesh

B.Tech Computer & Data Science (2023 – 2027)

LinkedIn | GitHub | brungishivaganesh@gmail.com



Education

STME, NMIMS University, Hyderabad **2023 - 2027**

B.Tech in Computer Science & Engineering (Data Science) — Data Structures & Algorithms, DBMS, Cloud Computing, Machine Learning and Full Stack Development.

Sri Guru Junior College, Ibrahimpatnam, Hyderabad **2021 - 2023**

Intermediate (MPC): 89% — strong analytical and mathematical foundation.

Chaitanya Central School (CBSE), Mahabubnagar **2013 - 2021**

Projects

Video Content Authenticator (VCA) | *FastAPI, OpenCV, SQLite, AWS EC2, Cognito* **2026 - Present**

- Built an end-to-end system to verify video authenticity using **SHA-256 hashing**, perceptual **dHash** analysis and **Ed25519 digital signatures**, ensuring integrity even across re-encoded media.
- Designed a **zero-trust architecture** with client-side key generation, integrated with a **FastAPI** backend.
- Deployed on **AWS EC2** with HTTPS (**Nginx, systemd**) and **AWS Cognito** auth; built a **Chrome extension** for real-time verification on social platforms (Instagram, LinkedIn).

Hackathon Platform – Serverless Cloud System | *React.js, Node.js, AWS* **2026**

- Built an event-driven backend using **AWS Lambda, API Gateway and DynamoDB** for scalable data handling and low-latency API operations.
- Implemented secure authentication using **AWS Cognito and JWT** with real-time dashboards for hackathon management.

Cloud-Based Student Result Management System | *AWS, React.js, Node.js* **2026**

- Built a cloud-based system to securely manage and access **student academic records and semester results**.
- Developed a **serverless backend (AWS Lambda, API Gateway, DynamoDB)** with **Cognito** role-based access for students and administrators; deployed on AWS.

AI-Powered Chest X-ray Disease Detection – Capgemini AI4 Health Hackathon (Rank 1) **2025**

- Trained a deep learning model in **Python** to detect diseases from chest X-rays and integrated the .h5 models into a **React + Node.js** app with probability scoring; **1st place among 19 teams** nationally.

Hybrid YOLO-LSTM Boxing Action Classifier – Personal Project **2026**

- Combined **YOLOv11 pose estimation** with **LSTM** sequence models to classify punches and blocks from live video (~98% validation accuracy); deployed as a **Flask** web app with **CUDA/TensorRT**-optimized inference.

Technical Skills

Languages: Python, C++, JavaScript, SQL

Core Concepts: Data Structures & Algorithms, Problem Solving & Logic Building, OOPs, DBMS, REST APIs

Cloud & DevOps: AWS (EC2, Lambda, API Gateway, DynamoDB, S3, RDS, Cognito), Google Cloud, Firebase, Docker, Nginx, systemd, Vercel, Render

Backend & Databases: FastAPI, Flask, Node.js, MySQL, PostgreSQL, SQLite, Firebase Firestore

Frontend & AI/ML: React, HTML, CSS | TensorFlow, PyTorch, Keras, OpenCV, YOLO (Ultralytics)

Tools: Git, GitHub, VS Code, Power BI, Excel

Certifications & Publications

Cloud: AWS Cloud Foundations – AWS | Google Cloud Foundations – GCP

Other: Microsoft Certifications | KL University Hackathon Certification | ICAT Certificate

Research Publication – IICTDS 2025: Co-authored and presented a paper on fruit & vegetable freshness detection using deep learning at the International Conference, NMIMS Chandigarh.

Achievements and Positions of Responsibility

- Winner – AI4 Health Hackathon (Judged by Capgemini)
- Research Paper Presenter – IICTDS 2025
- Organizer in Sports Club, NMIMS Hyderabad